

Definiteness, read from the signs

The quadratic form $\mathbf{x}^T A \mathbf{x}$ is classified entirely by the signs of the eigenvalues — not their sizes. Five classes, and the surface each one describes follows immediately.

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CLASSES

TYPE	CONDITION	EIGENVALUE SIGNS	SIGN OF THE FORM	LEVEL SURFACE	
ONE SIGN THROUGHOUT					
	Positive definite	all $\lambda > 0$	all strictly positive	> 0 for every $\mathbf{x} \neq \mathbf{0}$	ellipsoid
	Negative definite	all $\lambda < 0$	all strictly negative	< 0 for every $\mathbf{x} \neq \mathbf{0}$	inverted ellipsoid
ONE SIGN, WITH ZEROS					
	Positive semi-definite	all $\lambda \geq 0$, at least one $= 0$	non-negative, some zero	≥ 0 ; vanishes on the null space	degenerate ellipsoid — a cylinder
	Negative semi-definite	all $\lambda \leq 0$, at least one $= 0$	non-positive, some zero	≤ 0 ; vanishes on the null space	inverted cylinder
BOTH SIGNS PRESENT					
	Indefinite	some $\lambda > 0$ and some < 0	mixed	takes both signs	saddle — hyperboloid

In the eigenbasis the form becomes $\sum \lambda y_i^2$, and the sign question answers itself: a sum of squares with all-positive weights is positive, with mixed weights it takes both signs. Everything below is that substitution read off. In practice **Cholesky** tests definiteness faster than computing the eigenvalues does.